

100 Integrales, Indefinidas

Enumerados:

<https://drive.google.com/file/d/1TtOhEQCOYScnf9sef4R1tZK5F-ZWCOE1/view>

Comenzado el 9/10/24

terminado el 19/10/24 2:38

$$1) \int 5x^3 dx = \frac{5 \cdot x^4}{4} + c = \frac{5x^4}{4} + c$$

Elemental

$$\int x^a dx = \frac{x^{a+1}}{a+1} + c$$

$$2) \int (8x^2 - 5x^5) dx = \int 8x^2 dx - \int 5x^5 dx$$

$$\frac{8x^3}{3} - \frac{5x^6}{6} + c$$

8 y 5 son constantes
y se pueden sacar como
constante K

$$3) \int 3 dx = \int 3x^0 dx = 3 \int x^0 dx$$
$$3x + c$$

$$4) \int \left(\sqrt[3]{x} + \frac{5}{3} \right) dx = \int \sqrt[3]{x} + \int \frac{5}{3} dx$$

$$\int x^{\frac{1}{3}} + \frac{5}{3} \int x^0 dx$$

$$\frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} + \frac{5}{3}x + c = \frac{x^{\frac{4}{3}}}{\frac{4}{3}} + \frac{5}{3}x + c$$

$$5) \int \frac{1}{x^3} dx$$

Aca hay varios caminos.

1) tomo $\frac{1}{x^3} = x^{-3}$ y integro ∇

2) $\frac{P(x)}{q(x)}$ siendo $P(x) < q(x)$

$$\int x^{-3} dx = \frac{x^{-3+1}}{-3+1} + c = \frac{x^{-2}}{-2} + c = -\frac{1}{2x^2} + c$$

$$6) \int (2-x) \cdot \sqrt{x} dx$$

$$\int 2 \cdot \sqrt{x} - x \cdot \sqrt{x} dx = \int 2 \sqrt{x} dx - \int x \sqrt{x} dx$$

$$2 \int x^{\frac{1}{2}} dx - \int x' \cdot x^{\frac{1}{2}} dx$$

$$2 \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} - \int x^{\frac{3}{2}} dx$$

$$2 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} + c$$

$$1 + \frac{1}{2} = \frac{3}{2}$$

$$7) \int 2x \sqrt{1-3x^2} dx = 2 \int x \sqrt{1-3x^2} dx$$

Substitution

$$u = 1-3x^2$$

$$\frac{du}{dx} = -6x$$

$$du = -6x dx$$

$$-\frac{1}{6} du = x dx$$

$$= 2 \int -\sqrt{u} x dx = 2 \int -\sqrt{u} \cdot \frac{du}{-6}$$

$$= 2 \cdot \left(-\frac{1}{6}\right) \int \sqrt{u} du = -\frac{2}{6} \int u^{\frac{1}{2}} du$$

$$= -\frac{1}{3} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$= -\frac{1}{3} \cdot \frac{(1-3x^2)^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$8) \int \sqrt{5+x} dx = \int \sqrt{u} dx$$

$$u = 5+x$$

$$\frac{du}{dx} = 1$$

$$du = 1 dx$$

$$du = dx$$

$$= \int \sqrt{u} du$$

$$= \int u^{\frac{1}{2}} du$$

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{(5+x)^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$9) \int \frac{5}{x} dx = 5 \cdot \int \frac{1}{x} dx \\ = 5 \cdot \ln(|x|) + c$$

$$10) \int \frac{8 dx}{3x+2} = \int \frac{1}{3x+2} \cdot 8 \cdot dx \\ = 8 \cdot \int \frac{1}{3x+2} dx = 1 \cdot 2$$

1) $\frac{p(x)}{q(x)}$, teniendo $p(x) < q(x)$

2) Sustitución.

$$= 8 \int \frac{1}{3x+2} dx$$

$$= 8 \int \frac{1}{u}$$

$$= 8 \int \frac{1}{u} \cdot \frac{du}{3} = \frac{8}{3} \int \frac{1}{u} du$$

$$= \frac{8}{3} \cdot \ln(|u|) + c$$

$$= \frac{8 \ln(|3x+2|)}{3} + c$$

$$u = 3x+2$$

$$\frac{du}{dx} = 3x+2$$

$$du = 3 dx$$

$$\frac{du}{3} = dx$$

$$11) \int \frac{x^2+2x+1}{x^2-1} dx$$

$$\int \frac{(x+1)^2}{(x-1)(x+1)} dx$$

$$\int \frac{(x+1)}{(x-1)} dx = (x+1) = (x-1+2)$$

$$\int \frac{(x+2-1)}{(x-1)} dx = \int \frac{x-1}{x-1} + \int \frac{2}{x-1} dx$$

$$\frac{(x-1)}{x-1} + \frac{2}{x-1}$$

$$\int dx + 2 \ln(|x-1|) + c$$

$$x + 2 \ln(|x-1|) + c$$

$$\Rightarrow \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$x_1 = \frac{-2 \pm 0}{2} = -1$$

$$x_2 = \frac{-2 \pm 0}{2} = -1$$

$$12) \int \frac{x^2 + x - 2}{x-1} dx$$

$$\int \frac{(x-1)(x+2)}{(x-1)} dx$$

$$\frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot (-2)}}{2 \cdot 1} = x_1 = \frac{-1+3}{2} = \frac{2}{2} = 1$$

$$x_2 = \frac{-1-3}{2} = -2$$

$$\int \frac{(x-1)}{(x-1)} \cdot (x+2) dx$$

$$\int 1 \cdot (x+2) dx = \int 2 \int x dx$$

$$2 \cdot \frac{x^2}{2} + C$$

$$13) \int \frac{x^3 - 4x - 1}{x^2} dx = \int \frac{x^3}{x^2} dx + \int \frac{-4x}{x^2} dx + \int \frac{-1}{x^2} dx$$

✓ No puede ser $\ln(|x|)$, pues 12

$$= \int x dx - 4 \int \frac{x}{x^2} - 1 \int x^{-2} dx$$

$$= \frac{x^2}{2} - 4 \cdot \int \frac{1}{x} - 1 \cdot \frac{x^{-1}}{-1} + C$$

$$\frac{x^2}{2} - 4 \ln(|x|) + \frac{1}{x} + C$$

$$14) \int \frac{x^2 + 1}{x-1} dx$$

$$\int \frac{(x-1)(x+1)}{(x-1)} dx$$

$$\frac{(x+1)(x-1)}{x^2 + x - x - 1} \quad \times$$

$$\int 1 \cdot (x+1) dx = \int x dx + \int dx \quad \times$$

$$\frac{x^2}{2} + x + C$$

$$\int \frac{(x^2 + 1)}{(x-1)} dx$$

$$\int \frac{(x^2 - 1 + 2)}{x-1} dx = \int \frac{x^2 - 1}{x-1} + \frac{2}{x-1} = \int \frac{(x-1)(x+1)}{x-1} + 2 \int \frac{1}{x-1} dx$$

$$= \int \underline{x+1} dx + 2 \ln(|x-1|) + C$$

$$\frac{x^2}{2} + x + 2 \ln(|x-1|) + C$$

$$15) \int \frac{x^2 - x + 5}{x+3} dx$$

$$\begin{array}{r} x^2 - x + 5 \quad | \quad x+3 \\ \underline{-x^2 - 3x} \\ 4x + 5 \\ \underline{-4x - 12} \\ 17 \end{array}$$

$$D = d \cdot c + r$$

teo del cociente

$$\frac{D}{d} = \frac{d \cdot c + r}{d}$$

$$\frac{D}{d} = c + \frac{r}{d}$$

$$\int (x-1) + \frac{17}{x+3} dx$$

$$\int x - 4 + 17 \int \frac{1}{x+3} dx$$

$$\frac{x^2}{2} - 4x + 17 \ln|x+3| + c$$

$$16) \int (x^2 + 3x + 1)(2x+3) dx$$

Derivada

$$u = x^2 + 3x + 1$$

$$\frac{du}{dx} = x^2 + 3x + 1$$

$$du = 2x + 3 dx$$

$$\int u \cdot (2x+3) \cdot \frac{1}{2x+3} du$$

$$\int u \cdot du = \frac{u^2}{2} + c = \frac{(x^2 + 3x + 1)^2}{2} + c$$

$$17) \int \frac{x+1}{x+2} dx = \int \frac{x+2-1}{x+2} dx$$

$$\int \frac{x+2}{x+2} dx - \int \frac{1}{x+2} dx$$

$$x - \ln|x-2| + c$$

$$18) \int 7^{3x} dx$$

$$7 \cdot 7 = 49$$

Elemental

$$\int a^x = \frac{a^x}{\ln(a)} + c$$

$$\int 343^x dx$$

$$\frac{343^x}{\ln(343)} + c$$

Es válido. Aún también, hacer $u = 3x$

$$1d) \int e^{7x} dx =$$

$$\frac{e^{7x}}{7} + C$$

Elemental

$$\int e^{kx} dx = \frac{e^{kx}}{k} + C$$

$$20) \int (x^2 - 2)^4 \cdot x dx$$

$$u = x^2 - 2$$

$$\frac{du}{dx} = x^2 - 2$$

$$du = 2x dx$$

$$\underline{du = x dx}$$

$$\int u^4 \cdot x \cdot \frac{du}{x}$$

$$\int u^4 \cdot x \cdot \frac{1}{x} du$$

$$\int u^4 \cdot du =$$

$$\frac{u^5}{5} + C = \frac{(x^2 - 2)^5}{5} + C$$

$$21) \int \sqrt{3x-1} dx$$

$$u = 3x - 1$$

$$\frac{du}{dx} = 3x - 1$$

$$du = 3 dx$$

$$\frac{du}{3} = dx$$

$$\frac{1}{3} \int u^{\frac{1}{2}} du$$

$$\frac{1}{3} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{(3x-1)^{\frac{3}{2}}}{3 \cdot \frac{3}{2}} + C$$

$$22) \int x^2 \cdot e^{5x^3} dx$$

$$u = 5x^3$$

$$\frac{du}{dx} = 5x^3$$

$$du = 15x^2 dx$$

$$\frac{du}{15} = x^2 dx$$

$$\int e^{5x^3} \cdot x^2 dx$$

$$\int e^u \cdot \frac{du}{15}$$

$$\frac{1}{15} \cdot \int e^u$$

$$\frac{1}{15} \cdot e^u + C = \frac{e^{5x^3}}{15} + C$$

$$23) \int 3 \cos(3x) dx$$

$$3 \int \cos(3x) dx$$

$$3 \int \cos(u) \frac{du}{3}$$

$$\int \cos(u) = 1 \cdot \sin(u) + C = \sin(3x) + C$$

$$u = 3x$$

$$\frac{du}{dx} = 3$$

$$du = 3 dx$$

$$\frac{du}{3} = dx$$

$$24) \int \sin(2x+7) dx$$

$$\int \sin(u) \frac{du}{2}$$

$$\frac{1}{2} \int \sin(u)$$

$$-\frac{\cos(2x+7)}{2} + C$$

$$u = 2x+7$$

$$\frac{du}{dx} = 2$$

$$du = 2 dx$$

$$\frac{du}{2} = dx$$

$$25) \int x^3 \cdot \cos(x^4+1) dx$$

$$\int \cos(x^4+1) \cdot x^3 dx$$

$$\int \cos(u) \cdot \frac{du}{4} = \frac{1}{4} \sin(x^4+1) + C$$

$$u = x^4+1$$

$$\frac{du}{dx} = 4x^3$$

$$du = 4x^3 dx$$

$$26) \int (1+\cos(x))^2 \cdot \sin(x) dx$$

$$\int u^2 \cdot \sin(x) \cdot \frac{du}{- \sin(x)}$$

$$-\int u^2 du = -\frac{u^3}{3} + C = -\frac{(1+\cos(x))^3}{3} + C$$

$$u = 1+\cos(x)$$

$$\frac{du}{dx} = -\sin(x)$$

$$du = -\sin(x) dx$$

$$27) \int \frac{x dx}{\sqrt{1-x^2}}$$

$$u = 1-x^2$$

$$\frac{du}{dx} = -2x$$

$$du = -2x dx$$

$$-\frac{du}{2} = x dx$$

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{-2}$$

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{-2}$$

$$-\frac{1}{2} \int \frac{1}{\sqrt{u}} du = -\frac{1}{2} \int u^{-\frac{1}{2}} du = -\frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C = -\frac{(1-x^2)^{\frac{1}{2}}}{1} + C$$

$$28) \int \frac{x^2 + 2x}{(x+1)^2} dx$$

$$\int \frac{x^2 + 2x}{x^2 + 2x + 1} dx = \int \frac{x^2 + 2x + 1 - 1}{x^2 + 2x + 1} dx$$

$$\int dx - \int \frac{1}{(x+1)^2} dx$$

$$x - \int \frac{1}{u^2} du$$

$$x - \int u^{-2} du = x - \frac{u^{-1}}{-1} + C$$

$$= x - \frac{(x+1)^{-1}}{-1} + C$$

$$u = x+1$$

$$\frac{du}{dx} = \frac{1}{1} = 1$$

$$29) \int \sin^2(x) \cdot \cos(2x) dx$$

$$\int u^2 \cdot \frac{du}{2}$$

$$\frac{1}{2} \int u^2 du$$

$$\frac{1}{2} \cdot \frac{u^3}{3} + C = \frac{(\sin(2x))^3}{6} + C$$

$$u = \sin(2x)$$

$$\frac{du}{dx} = \sin(2x) \rightarrow f'(g) \cdot g'$$

$$du = \cos(2x) \cdot 2$$

$$\frac{du}{2} = \cos(2x)$$

$$30) \int \cos^2(x) \cdot \sin(x) dx$$

$$\int u^2 \cdot (-du)$$

$$-1 \frac{u^3}{3} + C = -\frac{(\cos(x))^3}{3} + C$$

$$u = \cos(x)$$

$$\frac{du}{dx} = -\cos(x)$$

$$du = -\sin(x) dx$$

$$-du = \sin(x) dx$$

$$31) \int \frac{1}{\cos(x)} \cdot \sin(x) dx = \int \frac{\sin(x)}{\cos(x)} dx$$

$$u = \cos(x)$$

$$\frac{du}{dx} = -\sin(x)$$

$$du = -\sin(x) dx$$

$$\int \frac{1}{\cos(x)} \cdot \sin(x) dx$$

$$\int \frac{-du}{u} = -\int \frac{du}{u} = -\ln(|\cos(x)|) + C$$

$$32) \int \frac{\sin(x)}{\cos^2(x)} dx$$

$$u = \cos(x)$$

$$\frac{du}{dx} = -\sin(x)$$

$$du = -\sin(x) dx$$

$$\int \frac{-1}{u^2} du$$

$$-\int u^{-2} du = -\frac{u^{-2+1}}{-2+1} + C$$

$$= \frac{-\cos(x)}{-1} + C$$

$$33) \int x \cdot \cot(x^2) dx$$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$\frac{du}{2} = x dx$$

$$\int \cot(x^2) \cdot x dx$$

$$\int \cot(u) \frac{du}{2}$$

$$\frac{1}{2} \int \cot(u) du = \frac{1}{2} \int \frac{\cos(u)}{\sin(u)} du$$

$$u = \sin(x)$$

$$\frac{du}{dx} = \cos(x)$$

$$du = \cos(x) dx$$

$$= \frac{1}{2} \int \frac{1}{\sin(u)} du$$

$$\frac{1}{2} \ln(|\sin(u)|) + C$$

$$\frac{1}{2} \ln(|\sin(x)|) + C$$

$$31) \int \sec(x) dx = \int \frac{1}{\cos(x)} du = \ln(|\cos(x)|) + C$$

$$35) \int (1 + \tan(x))^2 dx = \int (1 + 2\tan(x) + \tan^2(x)) dx$$

$$\int dx + 2 \int \tan(x) dx + \int \tan^2(x) dx$$

$$x + 2 \int \frac{\sin(x)}{\cos(x)} dx + \int \frac{\sin^2(x)}{\cos^2(x)} dx$$

$$\begin{aligned} u &= \cos \\ \frac{du}{dx} &= -\sin(x) \\ du &= -\sin(x) dx \end{aligned}$$

$$x + \int \frac{-1}{\cos(x)} dx + \int \frac{\sin^2(x)}{\cos^2(x)} dx$$

$$x - \ln(|\cos(x)|) + \int \frac{\sin^2(x)}{\cos^2(x)} dx$$

$$u = \cos(x)$$

$$\frac{du}{dx} = -\sin(x)$$

$$du = -\sin(x) dx$$

$$+ \int \left(\frac{\sin(x)}{\cos(x)} \right)^2 dx$$

$$+ \int \left(\frac{-1}{u} \right)^2 dx$$

$$+ \int \frac{1}{u^2} dx$$

$$u^{-1} = \frac{1}{u} \cdot -1$$

$$- \int u^{-2}$$

$$x - \ln(|\cos(x)|) - \frac{u^{-1}}{-1} + C = x - \ln(|\cos(x)|) + \cos(x) + C$$

$$36) \int \sec(\sqrt{x}) \cdot \frac{1}{\sqrt{x}} dx$$

$$u = x^{\frac{1}{2}}$$

$$\int \sec(x^{\frac{1}{2}}) \cdot \frac{1}{x^{\frac{1}{2}}} dx$$

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}} dx$$

$$\frac{du}{2} = \frac{1}{x^{\frac{1}{2}}}$$

$$\int \sec(u) \cdot 2 du$$

$$2 \int \frac{1}{\cos^2(u)} du = 2 \ln|\sec(u) + \tan(u)| + C$$

$$37) \int \sin^3(x) dx = \int \sin^2(x) \sin(x) dx$$

Sei:

$$\sin^2(x) + \cos^2(x) = 1$$

$$1 - \cos^2(x) = \sin^2(x)$$

$$= \int (1 - \cos^2(x)) \sin(x) dx$$

$$= \int \sin(x) dx - \int \cos^2(x) \sin(x) dx$$

$$\cos(x) - \int \cos^2(x) \sin(x) dx$$

$$u = \cos(x)$$

$$\frac{du}{dx} = -\sin(x)$$

$$du = -\sin(x) dx$$

$$\cos(x) - \int u^2 \cdot (-du)$$

$$\cos(x) + \frac{u^3}{3} + C = \cos(x) + \frac{\cos^3(x)}{3} + C$$

$$38) \int \sqrt{1 - \cos(x)} dx$$

$$1 - \cos(2x) = 2\sin^2(x) \rightarrow 1 - \cos x = 2\sin^2\left(\frac{x}{2}\right)$$

$$\int \sqrt{2\sin^2\left(\frac{x}{2}\right)} dx = \sqrt{2} \int \left|\sin\left(\frac{x}{2}\right)\right| dx$$

$$= \sqrt{2} \int \sin\left(\frac{x}{2}\right) du$$

$$u = \frac{x}{2}$$

$$\frac{du}{dx} = \frac{1}{2}$$

$$du = \frac{1}{2} dx$$

$$2du = dx$$

$$= \sqrt{2} \cdot 2 \int \cos(u) du$$

$$= 2\sqrt{2} \cos\left(\frac{x}{2}\right) + C$$

$$39) \int \cos^3 \frac{x}{3} dx = \int \cos^2 \left(\frac{x}{3} \right) \cdot \cos \left(\frac{x}{3} \right) dx$$

Se q:
 $\cos^2(x) = 1 - \sin^2(x)$

$$\int (1 - \sin^2 \left(\frac{x}{3} \right)) \cos \left(\frac{x}{3} \right) dx$$

$$\int \cos \left(\frac{x}{3} \right) - \sin^2 \left(\frac{x}{3} \right) \cdot \cos \left(\frac{x}{3} \right) dx$$

$$\int \cos \left(\frac{x}{3} \right) - \sin^2 \left(\frac{x}{3} \right) \cdot \cos \left(\frac{x}{3} \right) dx$$

$$= \sin \left(\frac{x}{3} \right) - \int u^2 \cdot du$$

$$= \sin \left(\frac{x}{3} \right) - 3 \int u^2 \cdot du$$

$$= \sin \left(\frac{x}{3} \right) - 3 \frac{u^3}{3} + c$$

$$= \sin \left(\frac{x}{3} \right) - \frac{3 \sin^3 \left(\frac{x}{3} \right)}{3} + c$$

$$u = \sin \left(\frac{x}{3} \right)$$

$$\frac{du}{dx} = \sin \left(\frac{x}{3} \right)$$

$$du = 3 \cos \left(\frac{x}{3} \right)$$

$$3 du = \dots$$

$$40) \int \frac{\ln(x)}{x} dx$$

$$\int \frac{1}{x} \ln(x) dx$$

$$\int u du$$

$$\frac{\ln^2(x)}{2} + c$$

$$u = \ln(x)$$

$$\frac{du}{dx} = \ln(x)$$

$$du = \frac{1}{x} dx$$

$$41) \int \frac{x dx}{\sqrt{x-1}} = \int \frac{x}{\sqrt{x-1}} dx$$

$$u^2 = 3x-1 \rightarrow x = \frac{u^2+1}{3}$$

$$2u du = 3 dx$$

$$= \int \frac{u^2+1}{\sqrt{u^2}} \cdot \frac{2u}{3} du$$

$$= \frac{2}{9} \int \frac{u^2+1}{u} du = \frac{2}{9} \int \frac{u^3+0}{u} du = \frac{2}{9} \int (u^2+1) du$$

$$\frac{2}{9} \left(\frac{u^3}{3} + u \right) + c = \frac{2}{9} \frac{(3x-1)^3}{3} + \frac{\sqrt{3x-1}}{9} + c$$

$$42) \int \frac{7 dx}{(3x+2)^4} = 7 \int \frac{1}{(3x+2)^4} dx =$$

$$u = 3x+2 \quad \frac{du}{dx} = 3 \quad \Rightarrow \int \frac{1 \cdot \frac{du}{3}}{u^4} = \frac{1}{3} \int u^{-4} du$$

$$\frac{du}{dx} = 3$$

$$du = 3 dx$$

$$\frac{du}{3} = dx$$

$$\frac{7(u)^{-3}}{3 \cdot (-3)} = \frac{7(3x+2)^{-3}}{-9} + C$$

$$43) \int \frac{1 - \ln(x)}{x \ln(x)} dx = \int \frac{1}{x \ln(x)} dx - \int \frac{\ln(x)}{x \ln(x)} dx$$

$$= \int \frac{1}{x} \cdot \frac{1}{\ln(x)} dx - \int \frac{1}{x} dx$$

$$u = \ln(x)$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$= \int \frac{1}{\ln(x)} \cdot du - \ln(|x|) + C$$

$$\ln(|\ln(x)|) - \ln(|x|) + C$$

$$44) \int \sin(x) \cdot e^{\cos(x)} dx$$

$$u = \cos(x)$$

$$\frac{du}{dx} = -\sin(x)$$

$$du = -\sin(x) dx$$

$$= \int e^u \cdot du = e^u + C$$

$$= e^{\cos(x)} + C$$

$$45) \int \frac{\cos(\ln(3x))}{x} dx = \int \cos(u) du$$

$$\sin(\ln(3x)) + C$$

$$u = \ln(3x)$$

$$\frac{du}{dx} = \frac{1}{3x} \cdot 3$$

46)

$$\tan^2 + 1 = \sec^2(x)$$

$$\int \sqrt{\tan^2(x) + 1} dx = \int \sec^2(x) dx$$

$$\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$$

$$47) \int \sec^2(5x) dx = \frac{1}{5} \int \sec^2(u) du \quad \begin{array}{l} u=5x \\ du=5dx \\ \frac{du}{5}=dx \end{array}$$

$$\frac{1}{5} \tan(u) + C = \frac{\tan(5x)}{5} + C$$

$$48) \int x \cdot \sin(x) dx$$

$$\rightarrow \int f \cdot g' = f \cdot g - \int f' \cdot g$$

$$x \cdot \cos(x) - \int 1 \cdot \cos(x) dx$$

$$-x \cos(x) - \int \cos(x) dx$$

$$-x \cos(x) + \sin(x) + C$$

$$\begin{array}{l} f=x, \quad f'(x)=1 \\ g'=\sin(x), \quad g=\cos(x) \end{array}$$

$$49) \int \ln(x) dx = x \ln(x) - x$$

$$50) \int \frac{x}{3} e^{2x} dx$$

$$\int \frac{1}{6} x e^u du$$

$$\frac{1}{6} \int \frac{u}{2} e^u du =$$

$$\frac{1}{2} \cdot \frac{1}{6} \cdot u \cdot e^u + \int e^u du$$

$$\frac{x}{6} \cdot e^u + e^u + C = \frac{x}{6} e^{2x} + e^{2x} + C$$

$$\begin{array}{l} u=2x \\ du=2dx \\ \frac{du}{2}=dx \end{array}$$

$$\begin{array}{l} f(x)=x, \quad f'(x)=1 \\ g'(x)=e^u, \quad g(x)=e^u \end{array}$$

$$\int f \cdot g' = f \cdot g - \int f' \cdot g$$

$$\begin{array}{l} u=2x \\ \frac{u}{2}=x \end{array}$$

$$51) \int x^4 \cdot \ln(x) dx$$

$$\int u dv = u \cdot v - \int v du$$

$$u = x^4 \quad dv = x^5$$

$$u = \ln(x) \quad dv = \frac{x^5}{x}$$

$$\ln(x) \cdot \frac{x^5}{5} - \int \frac{x^5}{5} \cdot \frac{1}{x} dx = \frac{x^5}{5} \ln(x) - \frac{x^5}{25} + C$$

Na e li Jo $\ln(x)$ Por q si no es Euclidean

$$52) \int 3x \cdot e^{-x^2} dx$$

$$u = -x^2$$

$$du = -2x dx$$

$$\frac{du}{2} = x dx$$

$$\int e^{-x^2} \cdot 3 \cdot x dx$$

$$\int e^u \cdot 3 \cdot \frac{du}{-2}$$

$$-\frac{3}{2} \int e^u du = -\frac{3}{2} e^u + C = -\frac{3}{2} e^{-x^2} + C$$

$$53) \int \frac{1}{e^x + 1} dx = \int \frac{\frac{1}{e^x}}{\frac{e^x + 1}{e^x}} dx = \int \frac{e^{-x}}{1 + e^x} dx$$

$$u = 1 + e^{-x}$$

$$du = -e^{-x} dx$$

$$\int \frac{1}{u} \cdot du = \ln(1 + e^{-x}) + C$$

$$54) \int \frac{1}{1 - \cos(x)} dx = \int \frac{1}{1 - \cos(x)} \cdot \frac{1 + \cos(x)}{1 + \cos(x)} dx$$

P. trigo

$$\sin^2(x) + \cos^2(x) = 1 \quad \int \frac{1 + \cos(x)}{1^2 - \cos^2(x)} dx = \int \frac{1 + \cos(x)}{\sin^2(x)}$$

$$\sin^2(x) = 1 - \cos^2(x) = \int \frac{1}{\sin^2(x)} dx + \int \frac{\cos(x)}{\sin^2(x)} dx$$

$$u = \sin$$

$$du = \cos(x)$$

$$= \int \csc^2 dx + \int \frac{1}{u^2} dx$$

$$-\cot(x) + \frac{u^{-1}}{-1} + C = -\cot(x) + \frac{\sin^{-1}}{-1} + C$$

$$55) \int \sec^3(x) = \int \sec x \cdot \sec^2(x)$$

$$\sec \cdot \cancel{\text{tg}} - \int \text{tg} \cdot \sec \cdot \cancel{\text{tg}} dx$$

$$\sec \cdot \text{tg} - \int \text{tg}^2 \sec dx$$

$$\sec \cdot \text{tg} - \int (1 - \sec^2) \sec dx$$

$$\sec \cdot \text{tg} + \int \sec^{\cancel{dx}} + \int \sec^3 dx$$

$$\sec \cdot \text{tg}(x) + \ln|\sec + \text{tg}| + C$$

$$56) \int (1 + \cos(x))^2 \sin(x) dx$$

$$\int u^2 du = \frac{u^3}{3} + C$$

$$= \frac{(1 + \cos(x))^3}{3}$$

$$\rightarrow \int u dv = u \cdot v - \int v du$$

$$u = \sec \quad du = \sec \cdot \text{tg}$$

$$dv = \sec^2 \quad v = \text{tg}$$

$$1 + \text{tg}^2 = \sec^2 dx$$

$$u = 1 + \cos(x)$$

$$du = -\sin(x) dx$$

$$56) \int \sin(x) \cdot \sec^2(x) dx = \int \sin(x) \cdot \frac{1}{\cos^2(x)} dx = \int \frac{\sin(x)}{\cos^2(x)} dx$$

$$- \int \frac{1}{u^2} du = - \frac{u^{-1}}{-1} + C$$

$$u = \cos^2(x)$$

$$du = -2 \cos(x) \sin(x) dx$$

$$= - \frac{(\cos(x))^{-1}}{-1} + C$$

$$58) \int x \operatorname{Arctg}(x) dx = \int u dv = u \cdot v - \int dv \cdot v$$

$$u = \operatorname{Arctg} \quad dv = \frac{1}{x^2+1} dx$$

$$du = \frac{1}{x^2+1} dx, \quad v = \frac{x}{2}$$

$$\operatorname{Arctg} \cdot \frac{x^2}{2} - \int \frac{1}{x^2+1} \cdot \frac{x^2}{2} dx$$

$$\operatorname{Arctg} \cdot \frac{x^2}{2} - \frac{1}{2} \int \frac{x^2}{x^2+1} dx$$

$$\sim -\frac{1}{2} \int \frac{x^2+1-1}{x^2+1} dx$$

$$-\frac{1}{2} \int dx - \int \frac{1}{x^2+1}$$

$$\operatorname{Arctg} \cdot \frac{x^2}{2} - \frac{1}{2} x - \frac{1}{2} \operatorname{Arctg} + C$$

$$59) \int \frac{\sin 2x + \cos 2x}{\sin 2x - \cos 2x} dx$$

$$u = \sin 2x - \cos 2x$$

$$du = 2 \cos 2x + 2 \sin 2x dx$$

$$\int \frac{1}{u} \cdot \frac{du}{2}$$

$$du = 2(\cos 2x + \sin 2x) dx$$

$$\frac{du}{2} = \cos 2x + \sin 2x$$

$$\frac{1}{2} \int \frac{1}{u} = \frac{1}{2} \ln(|u|) + C$$

$$\frac{\ln(\cos 2x + \sin 2x)}{2}$$

$$60) \int \frac{1}{x^2-1} dx = \int \frac{1}{(x+1)(x-1)} dx$$

$$\int \frac{A_1}{x-1} + \frac{A_2}{x+1} dx = \frac{1}{x^2-1} = \frac{A_1}{x-1} + \frac{A_2}{x+1}$$

$$\frac{1}{(x+1)(x-1)} = \frac{A_1(x-1) + A_2(x+1)}{(x+1)(x-1)}$$

$$\frac{1}{x^2-1} = \frac{A_1 x - A_1 + A_2 x + A_2}{x^2-1}$$

$$\frac{1}{x^2-1} = \frac{(A_1 + A_2)x + (-A_1 + A_2)}{x^2-1}$$

$$\begin{cases} A_1 + A_2 = 0 & A_1 = -A_2 & A_1 = -\frac{1}{2} \\ -A_1 + A_2 = 1 & 2A_2 = 1 & A_2 = \frac{1}{2} \end{cases}$$

$$= \frac{\frac{1}{2}}{x-1} + \frac{-\frac{1}{2}}{x+1}$$

$$\frac{1}{2} \int \frac{1}{x-1} dx - \frac{1}{2} \int \frac{1}{x+1} dx$$

$$\frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C$$

$$61) \int \frac{1}{\cos^2 \theta \sin^2 \theta} d\theta = \int \frac{1}{(\cos \cdot \sin)^2} d\theta$$

$$= \int \frac{1}{\left(\frac{\sin 2\theta}{2}\right)^2} d\theta = 4 \int \frac{1}{\sin^2 2\theta} d\theta$$

P.T.B

$$\sin 2x = 2 \sin x \cos x$$

$$\frac{\sin 2x}{2} = \sin x \cos x$$

$$u = 2x$$

$$du = 2 dx$$

$$= 4 \int \operatorname{cosec}^2 \theta d\theta =$$

$$= 4 \int \operatorname{cosec}^2 \left(\frac{u}{2}\right) \frac{du}{2}$$

$$2 \int \operatorname{cosec}^2(u) du$$

$$2(-\cot u) + C$$

$$-2 \cot(2x) + C$$

$$62) \int \frac{\sqrt{x} dx}{1+\sqrt{x}} = \int \frac{\sqrt{x}}{1+\sqrt{x}} \cdot \frac{1+\sqrt{x}}{1+\sqrt{x}} dx$$

$$= \int \frac{\sqrt{x} + x}{1+\sqrt{x}} dx$$

$$\frac{1+\sqrt{x}}{1+\sqrt{x}} \cdot \frac{1+\sqrt{x}}{1+\sqrt{x}}$$

$$= \frac{1+\sqrt{x}}{1+\sqrt{x}} \cdot x$$

$$= \int \frac{x + \sqrt{x}}{1+\sqrt{x}} dx = \int \frac{x}{1+\sqrt{x}} dx$$

$$\int \frac{u-2}{1+u} du = 2 \int \frac{(u-1)-1}{u+1} du = 2 \int \frac{u-1}{u+1} du$$

$$u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$dx = 2\sqrt{x} du$$

$$= x - 2\sqrt{x} + \ln(1+\sqrt{x})$$

$$2\sqrt{x} du = dx$$

$$2du = \frac{1}{\sqrt{x}} dx$$

$$63) \int \frac{1}{x \ln(x)} dx$$

$$u = \ln(x)$$

$$du = \frac{1}{x} dx$$

$$\frac{1}{x} du = \frac{1}{x} \cdot \frac{1}{x} dx$$

$$\int \frac{1}{u} du$$

$$\ln(|\ln(x)|) + C$$

$$64) \int (1-x^2) \sqrt{x} \sqrt{x} dx$$

$$\int (1-x^2) (x \cdot x^{\frac{1}{2}})^{\frac{1}{2}} dx$$

$$\int (1-x^2) (x^{\frac{3}{2}})^{\frac{1}{2}} dx = \int$$

$$\int (1-x^2) \cdot x^{\frac{3}{4}} dx$$

$$\int x^{\frac{3}{4}} - x^{-2} \cdot x^{\frac{3}{4}} dx$$

$$\int x^{\frac{3}{4}} - x^{\frac{5}{4}} dx = \int x^{\frac{3}{4}} dx - \int x^{\frac{5}{4}} dx$$

$$\frac{x^{\frac{3}{4}+1}}{\frac{3}{4}+1} - \frac{x^{\frac{5}{4}+1}}{\frac{5}{4}+1} + C$$

$$65) \int \frac{dx}{\sqrt{1-7x^2}}$$

$$\int \frac{1}{\sqrt{u}} \cdot \frac{du}{-14x}$$

$$-\frac{1}{14} \int u^{-1/2} \cdot \frac{du}{u}$$

$$-\frac{1}{14} \int \frac{u^{-1/2}}{u} du$$

$$-\frac{1}{14} \int u^{-3/2} du = -\frac{1}{14} \cdot \frac{u^{-1/2}}{-1/2} + C = -\frac{1}{14} \cdot \frac{(1-7x^2)^{-1/2}}{-1/2} + C = \frac{1}{7} \cdot \frac{1}{\sqrt{1-7x^2}} + C$$

$$u = 1-7x^2$$

$$du = -14x dx$$

$$\frac{du}{-14x} = dx$$

$$u = -14x$$

$$\frac{u}{-14} = x$$

$$66) \int \frac{x+2}{\sqrt{x^2+9}} = \int \frac{x+2}{\sqrt{x^2+3^2}} = \int \frac{x}{\sqrt{x^2+3^2}} dx + 2 \int \frac{1}{\sqrt{x^2+3^2}} dx$$

$$u = x^2 + 9$$

$$du = 2x dx$$

$$\frac{du}{2} = x dx$$

$$\frac{1}{2} \int \frac{1}{\sqrt{u}} du + 2 \ln(|x + \sqrt{x^2+3^2}|) + C$$

$$\frac{1}{2} \int u^{-\frac{1}{2}} du + 2 \ln(|x + \sqrt{x^2+3^2}|) + C$$

$$\frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + 2 \ln(|x + \sqrt{x^2+3^2}|) + C$$

$$\frac{1}{2} \frac{(x^2+9)^{\frac{1}{2}}}{\frac{1}{2}} + 2 \ln(|x + \sqrt{x^2+3^2}|) + C$$

$$67) \int \frac{(x+1)^2}{x^2+1} = \int \frac{(x+1)(x+1) dx}{x^2+1+2} = \int \frac{(x+1)(x+1)}{(x+1)(x-1)+2} dx$$

$$= \int \frac{x+1}{x-1+2} dx$$

Pol mi
tablic

$$= \int \frac{x+1}{x+1} dx = x + C$$

$$68) \int \frac{x}{1+x^4} \cdot dx$$

$$\int \frac{\frac{du}{2}}{1+u^2} = \frac{1}{2} \int \frac{1}{1+u^2} du$$

$$\frac{1}{2} \cdot \frac{1}{1} \operatorname{Arctg}\left(\frac{x^2}{1}\right) + C$$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$\frac{du}{2} = x dx$$

Elementar

$$\frac{1}{1^2+x^2} = \operatorname{Arctg}\left(\frac{x}{1}\right) + C$$

$$69) \int \frac{x^4}{x^2+1} dx = \int (x^2+1) - \frac{1}{x^2+1} dx$$

$$\begin{array}{r} x^4 \rightarrow \\ -x^2 \rightarrow \\ \hline x^2 + 1 \\ -x^2 - 1 \\ \hline 1 \end{array}$$

$$= \int x^2 dx + \int dx + \int -\frac{1}{x^2+1} dx$$

$$= \frac{x^3}{3} + x - \frac{1}{1} \operatorname{Arctg}\left(\frac{x}{1}\right) + C$$

$$70) \int \frac{1}{x^2+4x+5} dx = \int \frac{1}{(x+2)^2+1} dx \quad \begin{matrix} u = x+2 \\ du = dx \end{matrix}$$

$$\int \frac{1}{u^2+1} du = \frac{1}{1} + \operatorname{Arctg}\left(\frac{u}{1}\right) + c$$

$$\operatorname{Arctg}(x+2) + c$$

$$5895627701760545$$

$$71) \int \sqrt{36-x^2} dx = \int \sqrt{36-x^2} \cdot \frac{\sqrt{36-x^2}}{\sqrt{36-x^2}} dx$$

$$05/32 \quad 164$$

$$\int \frac{36-x^2}{\sqrt{36-x^2}} dx$$

$$= 36 \int \frac{1}{\sqrt{36-x^2}} - \int \frac{x^2}{\sqrt{36-x^2}} dx = \operatorname{Arcsen}\left(\frac{x}{6}\right) + \int \frac{1}{\sqrt{36-x^2}} x^2 dx$$

$$\int u dv = uv - \int v du$$

$$u = x^2 \quad du = 2x$$

$$dv = (36-x^2)^{-1/2} = \frac{1}{\sqrt{36-x^2}}$$

$$\Rightarrow x^2 \cdot (36-x^2)^{-1/2} = \int (36-x^2)^{-1/2} \cdot 2x dx$$

$$x^2 \cdot \sqrt{36-x^2} - \int (36-x^2)^{1/2} dx$$

$$x^2 \sqrt{36-x^2} - \int u^{1/2} x dx$$

$$\begin{matrix} u = 36 - x^2 \\ du = -2x dx \\ \frac{du}{-2x} = dx \end{matrix}$$

?? Nose

$$72) \int \sqrt{5-6x^2} dx = \int \sqrt{\left[\frac{5}{6} - x^2\right] 6} dx = \sqrt{6} \cdot \int \sqrt{\frac{5}{6} - x^2} dy$$

$$= \frac{1}{2} x \sqrt{\frac{5}{6} - x^2} + \frac{1}{2} \cdot \frac{5}{6} \operatorname{Arcsen}\left(\frac{x}{\sqrt{\frac{5}{6}}}\right) + c$$

$$\int \sqrt{a^2+x^2} dx =$$

$$\frac{1}{2} x \sqrt{a^2+x^2} + \frac{1}{2} a^2 \operatorname{Arcsen}\left(\frac{x}{a}\right) + c$$

$$73) \int \cos^3\left(\frac{x}{3}\right) dx = \int \cos^2\left(\frac{x}{3}\right) \cdot \cos\left(\frac{x}{3}\right) dx$$

$$= \int \left(1 - \sin^2\left(\frac{x}{3}\right)\right) \cdot \cos\left(\frac{x}{3}\right) dx$$

$$\cos^2 + \sin^2 = 1$$

$$\cos^2 = 1 - \sin^2$$

$$= \int \cos\left(\frac{x}{3}\right) - \sin^2\left(\frac{x}{3}\right) \cdot \cos\left(\frac{x}{3}\right) dx$$

$$\int f(g(x)) \cdot g'(x) dx \rightarrow \int \cos\left(\frac{1}{3}x\right) - \int \sin^2\left(\frac{x}{3}\right) \cdot \cos\left(\frac{x}{3}\right) dx$$

$$= 3 \sin\left(\frac{1}{3}x\right) - \int \sin^2\left(\frac{x}{3}\right) \cdot \cos\left(\frac{x}{3}\right) dx$$

$$\sim -\frac{1}{3} \int u^2 \cdot du$$

$$= 3 \sin\left(\frac{x}{3}\right) - \frac{1}{9} \cdot \frac{\left(\sin\left(\frac{x}{3}\right)\right)^3}{3} + C$$

$$u = \sin\left(\frac{x}{3}\right)$$

$$du = \frac{1}{3} \cos\left(\frac{x}{3}\right) dx$$

$$\rightarrow \frac{du}{3} = \cos\left(\frac{x}{3}\right) dx$$

$$74) \int \frac{2x+3}{x^2-5x+4} dx$$

$$\int \frac{2x+3}{(x-4)(x-1)} dx = \frac{A_1}{x-4} + \frac{A_2}{x-1}$$

$$= A_1(x-1) + A_2(x-4)$$

$$= A_1x - A_1 + A_2x - 4A_2$$

$$\begin{cases} A_1 + A_2 = 2 \\ -A_1 - 4A_2 = 3 \end{cases} \rightarrow A_1 = 2 - A_2 \rightarrow A_1 = 2 + \frac{5}{3} = \frac{11}{3}$$

$$+ A_2 - 4A_2 = 5$$

$$A_2 = -\frac{5}{3}$$

$$\int \frac{\frac{11}{3}}{x-4} dx + \int \frac{-\frac{5}{3}}{x-1} dx$$

$$\frac{11}{3} \int \frac{1}{x-4} - \frac{5}{3} \int \frac{1}{x-1} = \frac{11 \ln|x-4|}{3} + \frac{5 \ln|x-1|}{3}$$

$$\frac{5 \pm \sqrt{5^2 - 4 \cdot 1 \cdot 4}}{2 \cdot 1}$$

$$\frac{5 + \sqrt{9}}{2} = x_1 = \frac{5+3}{2} = \frac{8}{2} = 4$$

$$x_2 = \frac{5-3}{2} = \frac{2}{2} = 1$$

P. t. g

$$\sec^2 = \tan^2(x) + 1$$

$$75) \int \sec^4 x dx = \int \sec^2(x) \cdot \sec^2(x) dx$$

$$= \int (\tan^2(x) + 1) \cdot \sec^2(x) dx$$

$$\int (u^2 + 1) \cdot du$$

$$u = \tan$$

$$du = \sec^2 dx$$

$$\frac{u^3}{3} + u + C = \frac{\tan^3}{3} + \tan + C$$

$$76) \int \frac{5}{x^2 + 3x - 4} dx$$

$$\frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-4)}}{2 \cdot 1} = x_1 = \frac{-3 \pm 5}{2} = \frac{2}{2} = 1$$

$$x_2 = \frac{-3 - 5}{2} = \frac{-8}{2} = -4$$

$$5 \int \frac{1}{(x+1)(x-1)} dx = \frac{A_1}{x+1} + \frac{A_2}{(x-1)}$$

$$= \frac{A_1 x - A_1 + A_2 x + 4A_2}{(x+1)(x-1)}$$

$$\begin{cases} A_1 + A_2 = 0 \rightarrow A_1 = -A_2 \rightarrow A_1 = -\frac{1}{5} \\ -A_1 + 4A_2 = 1 \rightarrow 5A_2 = \frac{1}{5} \end{cases}$$

$$-\frac{1}{5} \int \frac{1}{x+1} dx + \frac{1}{5} \int \frac{1}{x-1} dx$$

$$-\frac{\ln(|x+1|)}{5} + \frac{\ln(|x-1|)}{5} + C$$

$$77) \int \frac{1}{x \sqrt{9x^2 - 25}} dx = \int \frac{1}{x \sqrt{(x^2 - \frac{25}{9})9}} = \frac{1}{3} \int \frac{1}{x \sqrt{x^2 - \frac{25}{9}}} dx$$

$$\frac{1}{x \sqrt{x^2 - \frac{25}{9}}} = \frac{1}{5} \operatorname{Arcsen}\left(\frac{x}{\frac{5}{3}}\right) + C$$

$$= \frac{1}{3} \int \frac{1}{x \sqrt{x^2 - \left(\frac{5}{3}\right)^2}} dx$$

$$\frac{1}{3} \left(\frac{1}{\frac{5}{3}} \right) \operatorname{Arcsen}\left(\frac{x}{\frac{5}{3}}\right) + C$$

$$78) \int \frac{1}{x^3 - 3x^2 + 2x} dx$$

$$\frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1} = x_1 = \frac{-3 - 1}{2} = -1$$

$$x_2 = \frac{-3 + 1}{2} = -1$$

$$\int \frac{1}{x(x^2 - 3x + 2)} dx$$

$$\int \frac{1}{x(x-1)(x-2)} dx = \frac{A_1}{x} + \frac{A_2}{x-1} + \frac{A_3}{x-2}$$

$$\frac{1}{x(x-1)(x-2)} = \frac{A_1(x-1)(x-2) + A_2(x)(x-2) + A_3(x)(x-1)}{(x)(x-1)(x-2)}$$

$$\frac{1}{x(x-1)(x-2)} = \frac{A_1(x^2 - 3x + 2) + A_2(x^2 - 2x) + A_3(x^2 - x)}{(x)(x-1)(x-2)}$$

$$\frac{1}{x(x-1)(x-2)} = \frac{A_1 x^2 - 3A_1 x + 2A_1 + A_2 x^2 - 2A_2 x + A_3 x^2 - A_3 x}{(x)(x-1)(x-2)}$$

$$\begin{cases} 2A_1 = 1 \rightarrow A_1 = \frac{1}{2} \\ -3A_1 - 2A_2 - A_3 = 0 \rightarrow -\frac{3}{2} - 2A_2 - A_3 = 0 \rightarrow -2A_2 - A_3 = \frac{3}{2} \rightarrow A_3 = -\frac{3}{2} - 2A_2 \\ A_1 + A_2 + A_3 = 0 \rightarrow \frac{1}{2} + A_2 - \frac{3}{2} - 2A_2 = 0 \\ \quad -\frac{2}{2} - A_2 = 0 \\ \quad -1 - A_2 = 0 \\ \quad -A_2 = 1 \\ \quad A_2 = -1 \\ \quad A_3 = \frac{1}{2} \end{cases}$$

$$\int \frac{\frac{1}{2}}{x} + \frac{-1}{x-1} + \frac{\frac{1}{2}}{x-2}$$

$$\frac{1}{2} \int \frac{1}{x} dx - \int \frac{1}{x-1} dx + \frac{1}{2} \int \frac{1}{x-2} dx$$

$$\frac{1}{2} \ln(|x|) - \ln(|x-1|) + \frac{1}{2} \ln(|x-2|) + C$$

Acá de go de hacer las del drive por el hecho de q es trigonometria muy compleja

7a) Resuelva la siguiente integral. Ayuda: Puede resultar más fácil hacer primero una sustitución y después resolver por partes.

$$\int x^3 \cos(x^2) dx$$

$$\int x^3 \cos(x^2) dx$$

$$\int x^2 \cos(x^2) x dx$$

$$\begin{aligned} u &= x^2 \\ du &= 2x dx \\ \frac{du}{2} &= x dx \end{aligned}$$

$$\int u \cos(u) \frac{du}{2}$$

$$\frac{1}{2} \int u \cdot \cos(u) du \rightarrow$$

$$\int u dv = uv - \int v du$$

$$\int u dv = uv - \int v du$$

$$u = u \quad du = 1$$

$$dv = \cos(u) \quad v = \sin(u)$$

$$u \sin(u) - \int \sin(u) \cdot 1 du$$

$$u \sin(u) + \cos(u) + c = x^2 \sin(x^2) + \cos(x^2) + c \quad \checkmark$$

8a)

$$\int e^{2x} dx = \frac{e^{2x}}{2} + c$$

$$\text{elemental } \int e^{kx} = \frac{e^{kx}}{k} + c$$

8b)

$$\int \frac{2x+3}{x^2+3x+4} dx$$

Notemos que:

$$f(x) = x^2 + 3x + 4 \rightarrow f'(x) = 2x + 3 \Rightarrow \frac{f'(x)}{f(x)}$$

Entonces:

$$\int \frac{2x+3}{x^2+3x+4} dx \Rightarrow \int \frac{du}{u}$$

$$\begin{aligned} du &= x^2 + 3x + 4 \\ du &= 2x + 3 dx \end{aligned}$$

$$= \ln|x^2+3x+4| + c$$

$$\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx \quad 82)$$

$$u = e^x + e^{-x}$$

$$du = e^x - e^{-x} dx$$

$$\int \frac{1}{u} du = \ln(|e^x + e^{-x}|) + C$$

$$\int \frac{\cos(x) - \sin(x)}{\cos(x) + \sin(x)} dx \quad 83)$$

$$u = \cos + \sin$$

$$du = -\sin + \cos$$

$$\int \frac{1}{u} du = \ln(|\cos(x) + \sin(x)|) + C$$

$$h) \int \frac{1}{\sin^2(x)} dx \quad 84)$$

$$u = \cot(x) = \frac{\cos}{\sin}$$

$$\frac{f}{g} = \frac{f' \cdot g - g' \cdot f}{g^2}$$

$$du = \frac{\cos}{\sin} dx$$

$$du = \frac{-\sin \cdot \sin + \cos \cdot \cos}{\sin^2}$$

$$du = \frac{-\sin^2 + \cos^2}{\sin^2} = -du = \frac{1}{\sin^2}$$

$$\cos^2 \cdot \sin^2 = 1$$

$$- \int du$$

$$-u = -\cot(x) + C$$

$$\int_3^5 \sqrt[3]{33-2x} dx \quad 85)$$

$$\int_3^5 (33-2x)^{\frac{1}{3}} dx = \frac{(33-2x)^{\frac{1}{3}+1}}{\frac{1}{3}+1} \Big|_3^5$$

$$= \frac{(33-2x)^{\frac{4}{3}}}{\frac{4}{3}} \Big|_3^5$$

$$= \frac{(33-10)^{\frac{4}{3}}}{\frac{4}{3}} - \frac{(33-6)^{\frac{4}{3}}}{\frac{4}{3}}$$

$$\int x e^x dx$$

06)

$$\int u dv = uv - \int v du$$

$$u = x \quad du = 1$$

$$v = e^x \quad dv = e^x$$

$$x e^x - \int e^x = x e^x - e^x + c$$

$$\int (1 - 2x) e^{-2x} dx$$

07)

$$u = -2x$$

$$du = -2 dx$$

$$\frac{du}{-2} = dx$$

$$\int e^{-2x} - 2x e^{-2x} dx = \int e^{-2x} - \int 2x e^{-2x} dx$$

$$\frac{e^{-2x}}{-2} - 2 \int x e^{-2x} dx$$

$$\frac{e^{-2x}}{-2} - 2 \left(x \frac{e^{-2x}}{-2} - \int 1 \cdot e^{-2x} \right)$$

$$\frac{e^{-2x}}{-2} - 2 \times \frac{e^{-2x}}{-2} - \frac{e^{-2x}}{-2} + c$$

$$\frac{e^{-2x}}{-2} - 2x + c$$

$$\int u dv = uv - \int v du$$

$$u = x \quad du = 1$$

$$v = e^{-2x} \quad dv = \frac{e^{-2x}}{-2}$$

$$\int \cos^4(x) dx$$

08)

Se q:

$$\cos^2 \cdot \sin^2(x) = 1$$

$$\cos^2 = \frac{1}{\sin^2(x)}$$

$$y \frac{1}{\sin^2} = \cos^2(x)$$

At
quedo

$$\frac{\cos^2}{\sin^2}$$

$$\int \cos^2(x) \cdot \cos^2(x) dx$$

$$\int \frac{1}{\sin^2(x)} \cdot \cos^2(x) dx$$

$$\int \frac{\cos^2(x)}{\sin^2(x)} dx = \int \cot^2(x) = \int (\csc^2 - 1) dx$$

$$\int \csc^2(x) - \int dx$$

$$-\cot(x) - x + c$$

$$b) \int \frac{1}{x \ln(x)} dx \quad 89)$$

$$u = x$$

$$\int \frac{1}{u} \cdot \frac{1}{u} dx$$

$$u = \ln(x)$$

$$du = \ln(x) dx$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\frac{du}{x} = dx$$

$$\int \frac{1}{u^2} dx = \int u^{-2} dx$$

$$\frac{u^{-1}}{-1} + c = \frac{x^{-1}}{-1} + c = -\frac{1}{x} + c$$

$$\int_2^4 \frac{x}{x^3 - 3x + 2} dx \quad 90)$$

Se me fue lo Putaloz y perdí

35.

esta quedaba:

$$\ln(|x-2|) + \ln(|x-1|) + c$$

$$c) \int x^2 \cos(x) dx \quad 91)$$

esta era una doble integral por partes y quedaba $\pm$

$$x^2 \sin(x) - 2 \sin(x) - \sin(x^2) + c \text{ Algo así}$$

$$\int_0^1 x^3 e^{x^2} dx \quad 92)$$

$$\int e^{x^2} \cdot x^2 \cdot x dx$$

$$u = x^2$$

$$du = 2x dx$$

$$\frac{du}{2} = x dx$$

$$\frac{1}{2} \int e^u \cdot u \cdot du$$

$$\rightarrow \int u dv = uv - \int v du$$

$$u = u \quad du = 1$$

$$dv = e^u \quad v = e^u$$

$$\frac{1}{2} (u \cdot e^u - \int e^u \cdot 1 dx)$$

$$\frac{1}{2} (ue^u - e^u + c)$$

$$\int \frac{x-1}{x^2+4} dx \quad 93)$$

$$\int \frac{x-1}{(x+2)(x+2)} dx = \frac{A_1}{(x+2)} + \frac{A_2}{(x+2)}$$

$$\frac{x-1}{(x+2)(x+2)} = \frac{A_1 x + 2A_1 + A_2 x + 2A_2}{(x+2)(x+2)}$$

$$\begin{cases} A_1 + A_2 = 1 \rightarrow A_1 = 1 - A_2 \rightarrow A_1 = \frac{1}{2} \\ 2A_1 + 2A_2 = -1 \quad 2(1 - A_2) + 2A_2 = -1 \end{cases}$$

$$\begin{aligned} 1 - A_2 + 2A_2 &= -\frac{1}{2} \\ 1 - A_2 &= -\frac{1}{2} \\ A_2 &= -\frac{1}{2} \end{aligned}$$

$$-\frac{1}{2} \int \frac{1}{x+2} dx + \frac{1}{2} \int \frac{1}{x+2} dx$$

$$= \frac{1}{2} \ln(|x+2|) + \frac{1}{2} \ln(|x+2|) + C$$

$$c) \int \frac{dx}{\sqrt{9-4x^2}} \quad 94)$$

$$\int \frac{1}{\sqrt{9-4x^2}} dx = \frac{1}{\sqrt{4}} \cdot \int \frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{4}} \cdot \text{Arcsen}(x) + C$$

$$\int (2x+1)(x^2+x+1) dx \quad 95)$$

$$\begin{aligned} u &= x^2 + x + 1 \\ du &= 2x + 1 dx \end{aligned}$$

$$\int (x^2+x+1) \cdot (2x+1) dx$$

$$\int u \cdot du = \frac{u^2}{2} + C = \frac{(x^2+x+1)^2}{2} + C$$

$$\int \frac{2x}{1+x^2} dx \quad 96)$$

$$\begin{aligned} u &= x^2 \\ du &= 2x dx \\ \frac{du}{2} &= x dx \end{aligned}$$

$$\int \frac{x}{1+x^2} dx$$

$$\frac{1}{2} \int \frac{1}{1+u} du = \frac{1}{2} \cdot \ln|1+u| + C = \frac{1}{2} \ln|1+x^2| + C$$

$$g) \int_0^2 x \ln(x^2 + 4) dx \quad 97)$$

$$\int \ln(x^2 + 4) \cdot x dx \quad \begin{matrix} u = x^2 \\ \frac{du}{2} = x dx \end{matrix}$$

$$\frac{1}{2} \int \ln(u+4) du = \frac{1}{2 \cdot (x^2 + 4)} + C$$

La 98 y 99 son la razón de por qué hice esto.

$$\int \sqrt{x} \ln(x) dx.$$

$$\int u dv = uv - \int v du$$

$$\begin{matrix} u = \ln(x) & du = \frac{1}{x} \\ dv = x^{1/2} & v = \frac{2}{3} x^{3/2} \end{matrix}$$

si has

$$u = \ln, \quad dv = x \ln(x) - x$$

yes onbude

$$\int x^{1/2} \ln(x) dx$$

$$= \ln(x) \cdot \frac{2}{3} x^{3/2} - \int \frac{2}{3} x^{3/2} \cdot \frac{1}{x} dx$$

$$\ln(x) \cdot \frac{2x^{3/2}}{3} - \frac{2}{3} \int \frac{x^{3/2}}{x} dx$$

$$\ln(x) \cdot \frac{2x^{3/2}}{3} - \frac{2}{3} \int x^{1/2} dx$$

$$\ln(x) \frac{2x^{3/2}}{3} - \frac{2}{3} \cdot \frac{x^{3/2}}{\frac{3}{2}} + C$$

$$\frac{3}{2} - 1 = \frac{1}{2}$$

$$\int \frac{1}{x^4} dx. \quad 99)$$

$$\int x^{-4} dx = \frac{x^{-3}}{-3} + C = \frac{1}{3x^3} + C$$

(100)

La integral 100 son los Amigos que hicimos en el camino!)